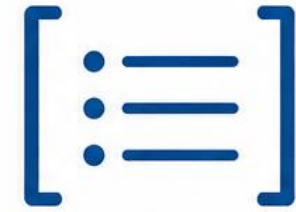


CONTENTS

1

LISTS

01



2

TUPLES

11



3

EXPERIMENT 4
MATERIALS DATA BASE PREPARATION
USING LISTS CONCEPT

21



PYTHON LIST

A list is an ordered, mutable collection of items.

SYNTAX

Lists are created using square brackets [].

Syntax:

```
list_name = [item1, item2, item3, ...]
```

Example:

```
fruits = ["Apple", "Banana", "Cherry"]
```

LIST INDEXING

	0	1	2	3	4
Index →	Apple	Banana	Cherry	Mango	Orange
Negative Index	-5	-4	-3	-2	-1

- Positive index starts from 0 (left to right).
- Negative index starts from -1 (right to left).
- Access element: `list_name[index]`

DESCRIPTION / KEY POINTS



1. ORDERED

Items in a list have a defined order. The order remains the same unless we change it.



2. MUTABLE (CHANGEABLE)

We can change, add or remove items after the list is created.



3. INDEXED

Each item has a unique index number starting from 0.



4. ALLOWS DUPLICATES

A list can contain duplicate items.



5. HETEROGENEOUS

A list can store items of different data types.



6. REPRESENTATION

Lists are written inside square brackets [] with items separated by commas.

EXAMPLES

```
numbers = [1, 2, 3, 4, 5]
```

List of integers

```
names = ["Amit", "Neha", "Rahul"]
```

List of strings

```
mixed = [1, "Hello", 3.14, True]
```

List with different data types

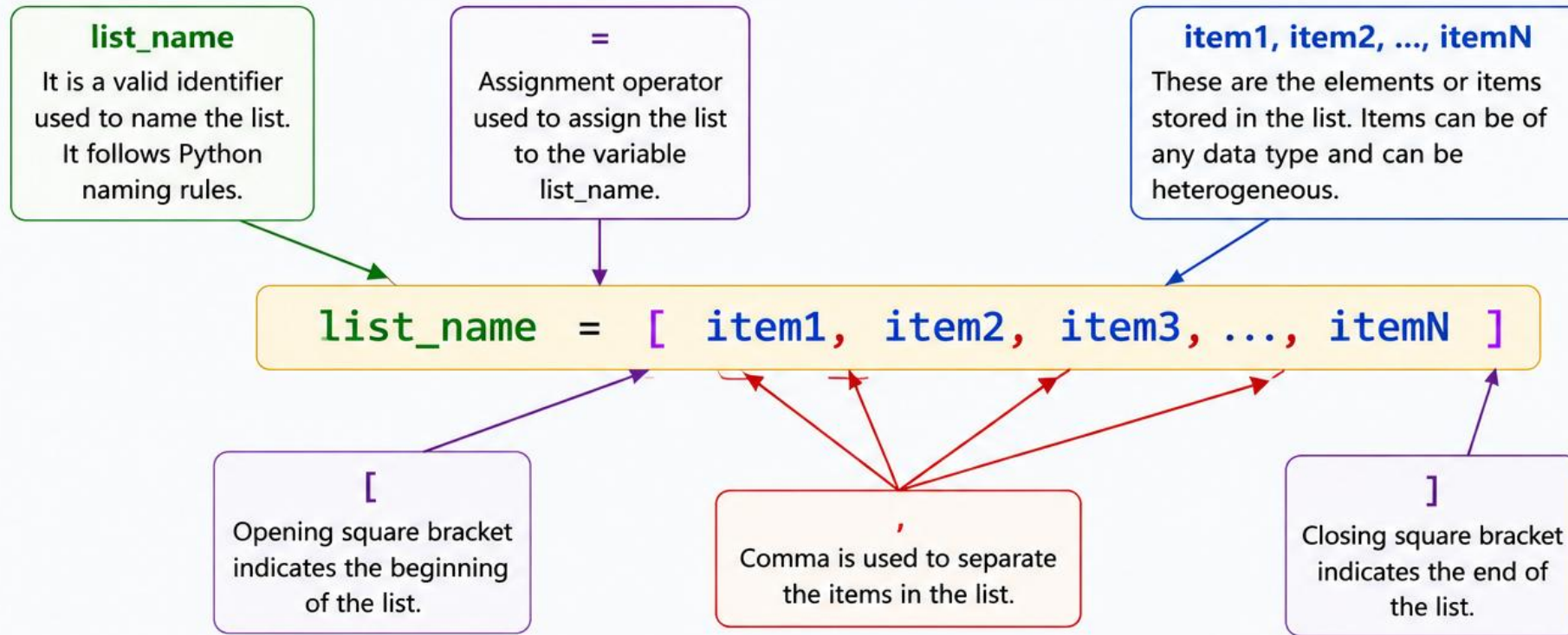
```
empty_list = []
```

Empty list

```
duplicates = [1, 2, 2, 3, 1]
```

List with duplicates

PYTHON LIST – SYNTAX



LIST INDEXING (Starts from 0)



- **Positive index** starts from 0 on the left and increases by 1 towards the right.
- **Negative index** starts from -1 on the right and decreases by 1 towards the left.

Example

```
marks = [85, 92, 78, 90]
marks[0]  -> 85
marks[2]  -> 78
marks[-1] -> 90
marks[-2] -> 78
```

PYTHON TUPLE

A tuple is an **ordered**, **immutable** collection of items.

SYNTAX

Tuples are created using parentheses ().

```
tuple_name = (item1, item2, item3, ...)
```

EXAMPLE

```
student = ("Amit", 20, "CSE", 9.2)
```

TUPLE INDEXING

	0	1	2	3
Index →	Amit	20	CSE	9.2
Negative Index →	-4	-3	-2	-1

- Positive index starts from 0 (left to right).
- Negative index starts from -1 (right to left).
- Access element: `tuple_name[index]`
- Example: `student[2]` → "CSE"

DESCRIPTION / KEY POINTS



1. ORDERED

Items in a tuple have a defined order.
The order remains the same unless we recreate it.



2. IMMUTABLE (READ-ONLY)

Once a tuple is created, we cannot change, add or remove its items.



3. INDEXED

Each item has a unique index number starting from 0.



4. ALLOWS DUPLICATES

A tuple can contain duplicate items.



5. HETEROGENEOUS

A tuple can store items of different data types.



6. REPRESENTATION

Tuples are written inside parentheses () with items separated by commas.

EXAMPLES

```
numbers = (1, 2, 3, 4, 5)
```

Tuple of integers

```
names = ("Amit", "Neha", "Rahul")
```

Tuple of strings

```
mixed = (1, "Hello", 3.14, True)
```

Tuple with different data types

```
single_item = (10,)
```

Tuple with single item

```
duplicates = (1, 2, 2, 3, 1)
```

Tuple with duplicates

FLOWCHART: CLASSIFICATION OF STEEL GRADES USING LISTS

